Embedded Systems I

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2020.09.01

- ✓ Course Title: Embedded Systems I
- ✓ Fall 2020 INC4065-01 (02)
- ✓ MW 13:00~15:00/TTH 13:00~15:00
- ✓ Lecturer: Professor Donald Kim (Tel: 2260-3234)
- ✓ Office Hour: By Appointment

- ✓ **Text: Operating System Design: The Xinu Approach Linksys Version**
- ✓ **Prerequisites: C Language, Microprocessor Programming**

Course Outline: **Few operating system courses explain underlying patterns and principles that form the basis for operating system construction. Students form the impression that an operating system is a blackbox. They only learn how an operating system appears from the outside, namely a set of interface functions, without knowing the internal structures. When working in the embedded world, knowledge of principles and structures is essential for building new mechanisms inside an operating system or porting it on new hardware. Furthermore, writing applications for embedded devices requires an appreciation for the underlying operating system. This course will review the major system components, and impose a hierarchical design paradigm that organizes the components in an orderly, understandable manner. It will guide them through the construction of a conventional process-oriented operating system, using practical, straightforward primitives. The Blockchain technology is the most demand skill in the financial institutions in these days. After the midterm, we focus on how to design Distributed Applications on the Ethereum platform. We cover the DAPP design environment including Truffle, Metamask, Web3, and Ganache. Solidity programming language is introduced and a DAPP project is given for implementing an payment application.**

- ✓ **Grading: 30% Midterm, 35%Final Exam, 10% Homework, 15% Lab Assignments, 10% Attendance**

- ✓ **Grading Scale: 30% A, 40% B, 30% C or below C**

- ✓ **Note: Those who miss the classes more than 7 times will receive F. One absent will deduct 10% of attendance score. Being late for three times is equivalent to one absence .**

✓ Agenda

0	주차(Week)	강의내용(Class Topic & Contents)	강의활동유형(Class Type)
0	1	(Lecture) Linux and Xinu Internals 1	강의+실습
0	2	(Lecture) Linux and Xinu Internals 2	강의+실습
0	3	(Lecture) Linux and Xinu Internals 3	강의+실습
0	4	(Lab) Linux and Xinu Internals Lab	강의+실습
0	5	(Lecture) Socket Programming 1	강의+실습
0	6	(Lecture) Socket Programming 2	강의+실습
0	7	(Lab) Socket Programming Lab	강의+실습
0	8	Midterm Examination	강의+시험
0	9	(Lecture) Data Models, Relational Database System, Object-Oriented Database System	강의+실습
0	10	(Lecture) Introduction to Blockchain (Ethereum Platform)	강의+실습
0	11	(Lecture) Solidity Programming Language	강의+실습
0	12	(Lecture) Truffle, Metamask, Web3, Ganesh	강의+실습
0	13	(Lecture) HTML, CSS	강의+실습
0	14	(Lab) Blockchain Projects	강의+실습
0	15	Final examination	강의+실습

Note: Lectures and lab assignments can be changed as we go through the course.